

2 Chloroplasts and mitochondria are two organelles found in some plant cells.

(a) Compare and contrast the structure of a chloroplast with the structure of a mitochondrion.

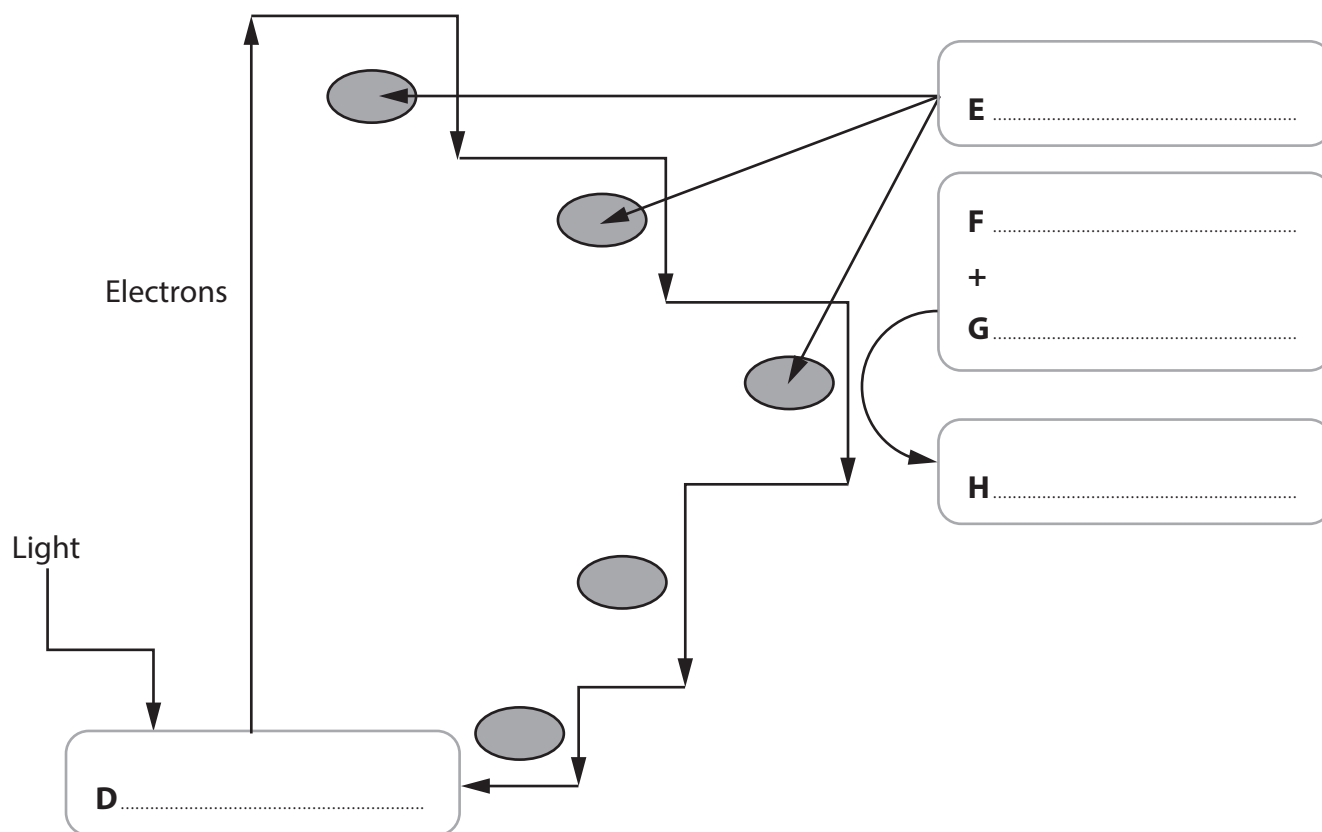
(3)

(b) The light-dependent reactions of photosynthesis can result in either cyclic or non-cyclic photophosphorylation.

(i) The diagram shows cyclic photophosphorylation.

Complete the diagram by writing the correct word or words on the dotted lines, labelled D, E, F, G and H.

(3)



(ii) Name **one** molecule that is produced in non-cyclic photophosphorylation that is **not** produced in cyclic photophosphorylation.

(1)

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(iii) Describe the role of photolysis in non-cyclic photophosphorylation.

(2)

(Total for Question 2 = 9 marks)
